Prabjot Singh

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Education

New York Institute of Technology Expected May 2027
B.S. Computer Science, GPA: 3.82 *Old Westbury, NY*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Organization & Architecture, Computer Networks, Database Management, Software Engineering, Artificial Intelligence

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, Node.js, FastAPI, LangChain, Pydantic

Tools & Technologies: Git, CI/CD, PostgreSQL, MySQL, OracleDB, AWS, Supabase

Technical Experience

Optimum Aug 2026 – Present
OSS/BSS Development Intern (Software Engineering) *Bethpage, NY*

- Replaced Excel-based modernization tracking with a full-stack platform **(React, TypeScript, Node.js, PostgreSQL)** for **140 applications** and **100+ users**, reducing time spent on manual tracking by **60%** through Jira automation.
- Implemented container-based connectivity to lab services across separate networks for an **MCP server** automating QA workflows, including provisioning and testing, saving the team an estimated **30 hours weekly**.
- Eliminated the department's recurring manual booking process by building a **Playwright-based** reservation tool integrated with **Grok Bot**, an AI agent platform, to automatically reserve workspaces each week.

Axentra OS April 2026 – Aug 2026
Software Engineer Intern *Remote*

- Architected a Shopify OAuth 2.0 pipeline using **FastAPI** and **ngrok**, handling a 10-step merchant authorization flow with HMAC verification, achieving **100%** token accuracy with tokens secured in **PostgreSQL** via **SQLAlchemy**.
- Implemented a warranty reporting layer consolidating store and third-party invoice registrations across **2** purchase workflows, enabling standard and extended warranty claim eligibility tracking.
- Developed reusable CRUD modules in **Python** for a **36-app** Shopify operating system, enforcing async PostgreSQL operations, request validation, and API response handling across core commerce workflows.

New York Institute of Technology June 2025 – Aug 2025
Robotics Engineering Assistant *Old Westbury, NY*

- Executed and analyzed **Python** motion-control scripts on a **Raspberry Pi** robot across **200+ physical runs**, diagnosing directional movement and hardware-software integration issues.
- Reduced cumulative positional drift by **20%** through iterative motion-control tuning and repeated physical testing.

Entrepreneurship and Technology Innovation Center (NYIT) April 2025 – June 2025
Software Engineering Fellow *Old Westbury, NY*

- Built a unified database management application supporting **PostgreSQL**, **MySQL**, and **Oracle**, combining schema inspection and GUI-based CRUD operations to reduce workflow setup time by nearly **50%**.
- Designed a reusable JDBC layer with shared connection and routing logic across **12 database operations**, eliminating **150+ lines** of duplicated code and reducing connection overhead by **20%**.

Projects

Lumen Finance: Personal Finance App | *React Native, Expo, Supabase, TypeScript, RESTful APIs* GitHub

- Spearheaded a cross-platform personal finance app in **React Native** and **Expo** with **Supabase** authentication, JWT sessions, and Row Level Security, supporting secure finance tracking across **150+** tested user accounts.
- Integrated Meta **Llama 3.3 70B** via **Groq** and Supabase Edge Functions to generate personalized finance insights.
- Developed admin tools for real-time user search, role assignment, and management of over **1,200** transaction records.

ScholarAI: AI Research Assistant | *Python, Pydantic, OpenAI, Anthropic, LangChain* GitHub

- Developed a **LangChain** AI agent pipeline in **Python** orchestrating **3** specialized tools to generate structured, multi-section research reports with summaries, citations, and tool-based reasoning.
- Validated research report outputs using **Pydantic** schemas to maintain consistent JSON formatting.
- Configured dual-model support with **GPT-4o mini** and **Claude 3.5 Sonnet**, adding fallback on parsing failures.